NUMBER SYSTEMS

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- Review the decimal number system
- Count in the binary number system
- Convert from decimal to binary and from binary to decimal
- Apply arithmetic operations to binary numbers
- Hexadecimal system
- Octal system
- Determine the 1's and 2's complements of a binary number
- Express signed binary numbers in sign-magnitude
- Express signed binary numbers in1's complement
- Express signed binary numbers in 2's complement
- Carry out arithmetic operations with signed binary numbers
- Convert between the binary and hexadecimal number systems
- Add numbers in hexadecimal form
- Convert between the binary and octal number systems

- Decimal Numbers
- Binary Numbers
- Decimal-to-Binary Conversion
- Binary Arithmetic
- Complements of Binary Numbers Signed Numbers
- Arithmetic Operations with Signed Numbers
- Hexadecimal Numbers
- Octal Numbers

Decimal Numbers

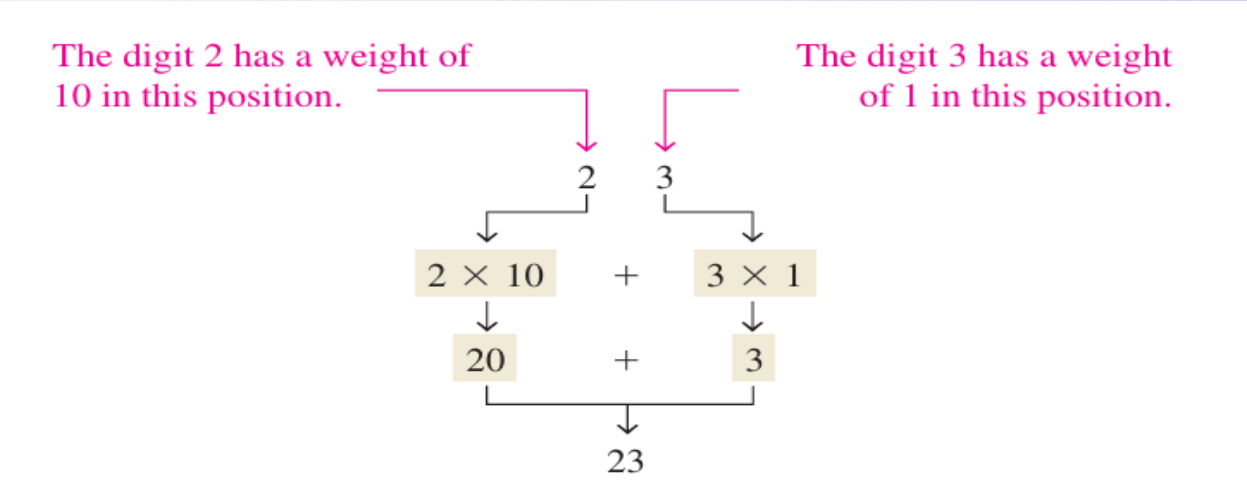
In the decimal number system each of the ten digits, 0 through 9, represents a certain quantity.

The position of each digit in a decimal number indicates the magnitude of the quantity represented and can be assigned a weight.

$$\ldots\ 10^5\ 10^4\ 10^3\ 10^2\ 10^1\ 10^0$$

$$10^2\ 10^1\ 10^0.10^{-1}\ 10^{-2}\ 10^{-3}\ \ldots$$

Decimal point



Decimal Numbers

Express the decimal number 47 as a sum of the values of each digit.

Solution

The digit 4 has a weight of 10, which is 10^1, as indicated by its position. The digit 7 has a weight of 1, which is 10^0, as indicated by its position.

$$\begin{aligned} 47 &= (4 \times 10^1) + (7 \times 10^0) \\ &= (4 \times 10) + (7 \times 1) = \mathbf{40 + 7} \end{aligned}$$

Express the decimal number 568.23 as a sum of the values of each digit.

Solution

The whole number digit 5 has a weight of 100, which is 10^2, the digit 6 has a weight of 10, which is 10^1, the digit 8 has a weight of 1, which is 10^0, the fractional digit 2 has a weight of 0.1, which is 10^{-1}, and the fractional digit 3 has a weight of 0.01, which is 10^{-2}.

$$\begin{aligned} 568.23 &= (5 \times 10^2) + (6 \times 10^1) + (8 \times 10^0) + (2 \times 10^{-1}) + (3 \times 10^{-2}) \\ &= (5 \times 100) + (6 \times 10) + (8 \times 1) + (2 \times 0.1) + (3 \times 0.01) \\ &= \mathbf{500 + 60 + 8 + 0.2 + 0.03} \end{aligned}$$

Binary Numbers

A binary number is a weighted number. The right-most bit is the LSB (least significant bit) in a binary whole number and has a weight of 20 = 1. The weights increase from right to left by a power of two for each bit. The left-most bit is the MSB (most significant bit); its weight depends on the size of the binary number.

The weight structure of a binary number is

$$2^{n-1} \ldots 2^3\, 2^2\, 2^1\, 2^0 \,.\, 2^{-1}\, 2^{-2} \ldots 2^{-n}$$

Binary point

Binary weights.

Positive Powers of Two (Whole Numbers)									Negative Powers of Two (Fractional Number)					
2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}	2^{-4}	2^{-5}	2^{-6}
256	128	64	32	16	8	4	2	1	1/2 0.5	1/4 0.25	1/8 0.125	1/16 0.625	1/32 0.03125	1/64 0.015625

Binary Numbers

Convert the binary whole number 1101101 to decimal.

Solution

Determine the weight of each bit that is a 1, and then find the sum of the weights to get the decimal number.

$$\begin{array}{rl} \text{Weight:} & 2^6\ 2^5\ 2^4\ 2^3\ 2^2\ 2^1\ 2^0 \\ \text{Binary number:} & 1\ \ 1\ \ 0\ \ 1\ \ 1\ \ 0\ \ 1 \end{array}$$

$$\begin{aligned} 1101101 &= 2^6 + 2^5 + 2^3 + 2^2 + 2^0 \\ &= 64 + 32 + 8 + 4 + 1 = \mathbf{109} \end{aligned}$$

Convert the fractional binary number 0.1011 to decimal.

Solution

Determine the weight of each bit that is a 1, and then sum the weights to get the decimal fraction.

$$\begin{array}{rl} \text{Weight:} & \quad\ \ 2^{-1}\quad 2^{-2}\quad 2^{-3}\quad 2^{-4} \\ \text{Binary number:} & 0\ .\ 1\quad\ \ 0\quad\ \ 1\quad\ \ 1 \end{array}$$

$$\begin{aligned} 0.1011 &= 2^{-1} + 2^{-3} + 2^{-4} \\ &= 0.5 + 0.125 + 0.0625 = \mathbf{0.6875} \end{aligned}$$

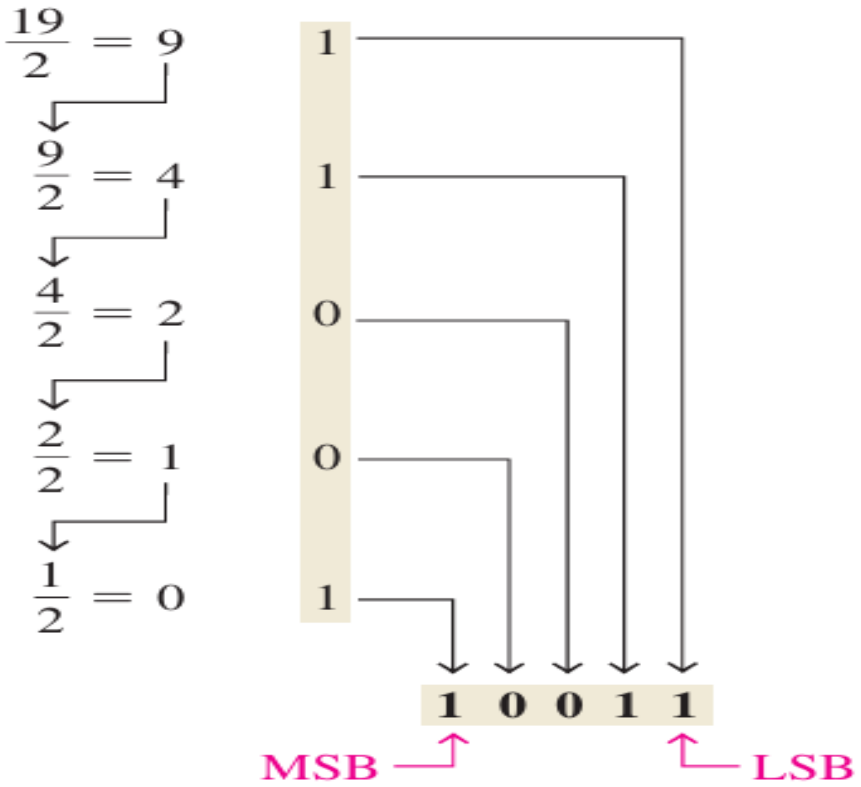
Decimal-to-Binary Conversion

Convert the following decimal numbers to binary:

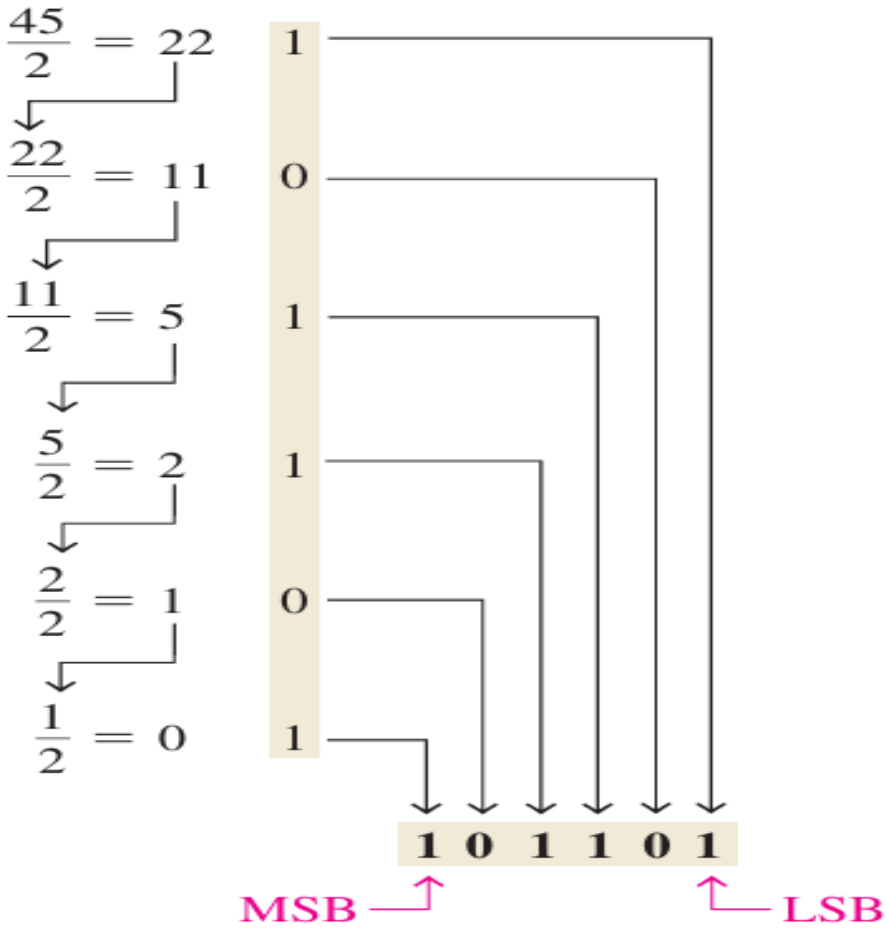
(a) 19 **(b)** 45

Solution

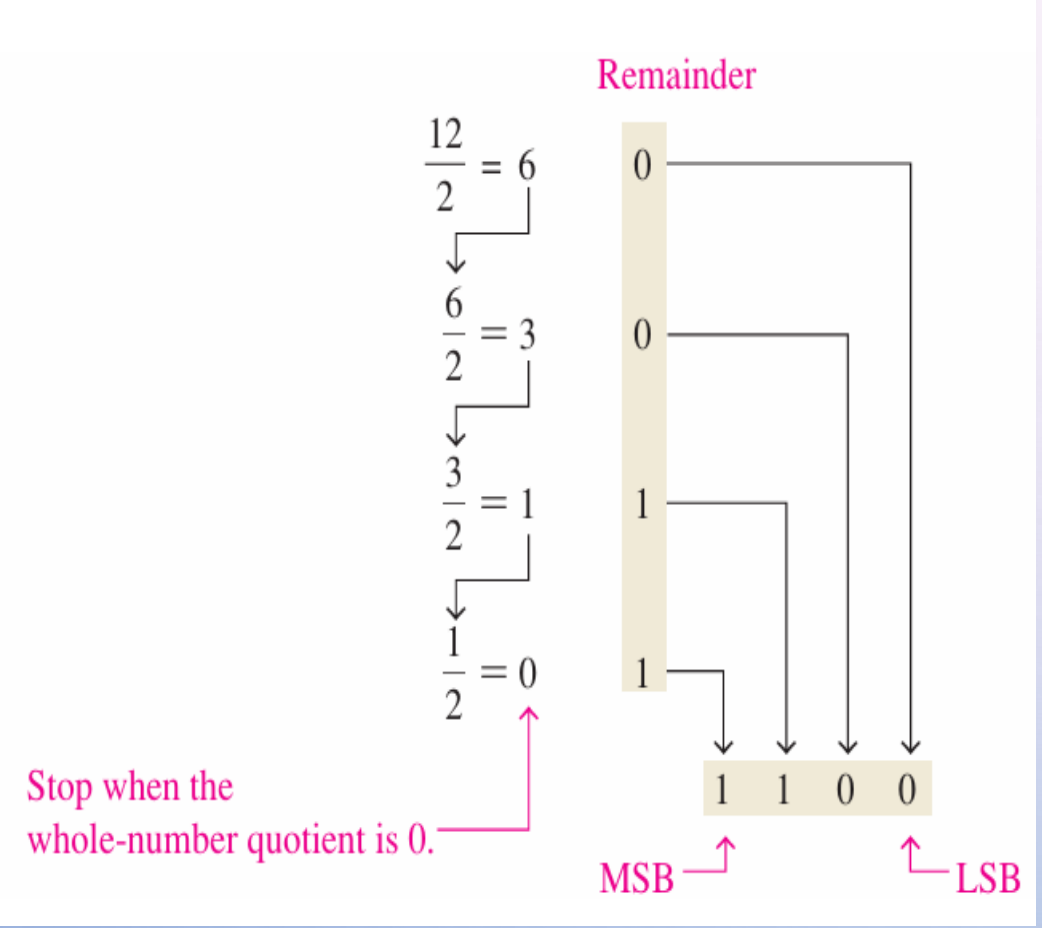
(a)

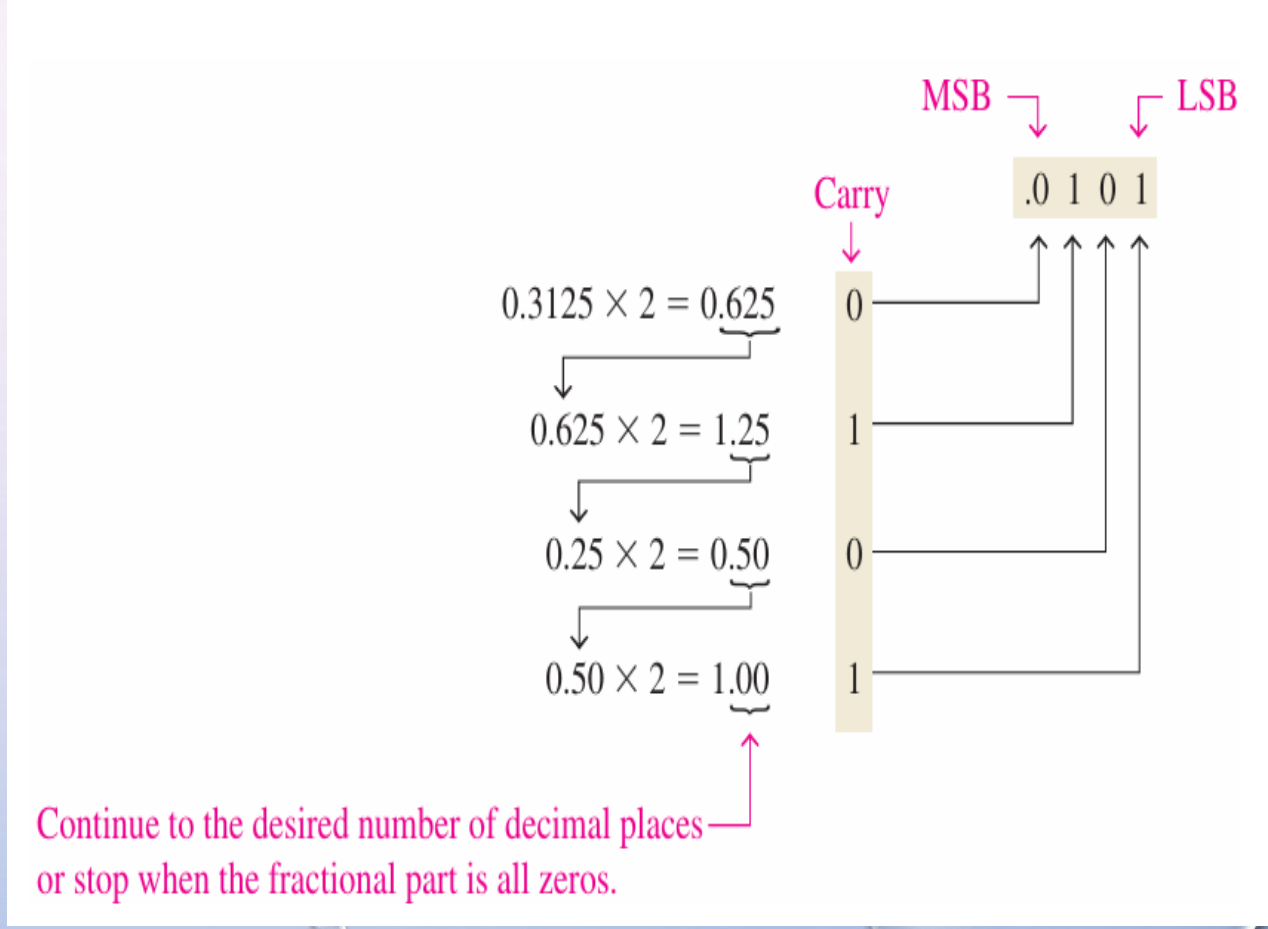


(b)



Decimal-to-Binary Conversion





(12.3125) =(1100.0101)

Decimal-to-Binary Conversion

11. Convert each decimal number to binary by using the sum-of-weights method:

(a) 12 **(b)** 15 **(c)** 25 **(d)** 50
(e) 65 **(f)** 97 **(g)** 127 **(h)** 198

12. Convert each decimal fraction to binary using the sum-of-weights method:

(a) 0.26 **(b)** 0.762 **(c)** 0.0975

13. Convert each decimal number to binary using repeated division by 2:

(a) 13 **(b)** 17 **(c)** 23 **(d)** 30
(e) 35 **(f)** 40 **(g)** 49 **(h)** 60

14. Convert each decimal fraction to binary using repeated multiplication by 2:

(a) 0.76 **(b)** 0.456 **(c)** 0.8732

Binary Arithmetic

After completing this section, you should be able to

- Add binary numbers
- Subtract binary numbers
- Multiply binary numbers
- Divide binary numbers

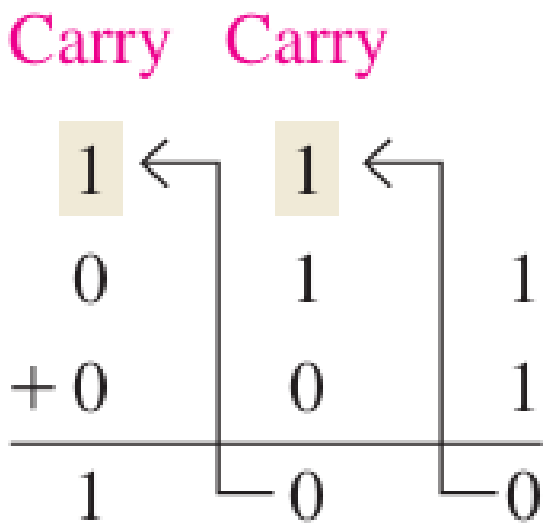
Binary Addition

The four basic rules for adding binary digits (bits) are as follows:

$0 + 0 = 0$	Sum of 0 with a carry of 0
$0 + 1 = 1$	Sum of 1 with a carry of 0
$1 + 0 = 1$	Sum of 1 with a carry of 0
$1 + 1 = 10$	Sum of 0 with a carry of 1

In binary $1 + 1 = 10$, not 2.

Binary Addition



1	+ 0 + 0 = 01	Sum of 1 with a carry of 0
1	+ 1 + 0 = 10	Sum of 0 with a carry of 1
1	+ 0 + 1 = 10	Sum of 0 with a carry of 1
1	+ 1 + 1 = 11	Sum of 1 with a carry of 1

Binary Addition

Add the following binary numbers:

(a) 11 + 11 (b) 100 + 10

(c) 111 + 11 (d) 110 + 100

Solution

The equivalent decimal addition is also shown for reference.

(a)
$$\begin{array}{r} 11 \\ +\ 11 \\ \hline \mathbf{110} \end{array} \qquad \begin{array}{r} 3 \\ +\ 3 \\ \hline 6 \end{array}$$

(b)
$$\begin{array}{r} 100 \\ +\ 10 \\ \hline \mathbf{110} \end{array} \qquad \begin{array}{r} 4 \\ +\ 2 \\ \hline 6 \end{array}$$

(c)
$$\begin{array}{r} 111 \\ +\ 11 \\ \hline \mathbf{1010} \end{array} \qquad \begin{array}{r} 7 \\ +\ 3 \\ \hline 10 \end{array}$$

(d)
$$\begin{array}{r} 110 \\ +\ 100 \\ \hline \mathbf{1010} \end{array} \qquad \begin{array}{r} 6 \\ +\ 4 \\ \hline 10 \end{array}$$

Binary Subtraction

The four basic rules for subtracting bits are as follows:

$$0 - 0 = 0$$
$$1 - 1 = 0$$
$$1 - 0 = 1$$
$$10 - 1 = 1 \qquad 0 - 1 \text{ with a borrow of } 1$$

Perform the following binary subtractions:

(a) 11 − 01 **(b)** 11 − 10

Solution

(a)

$$\begin{array}{r} 11 \\ -01 \\ \hline \mathbf{10} \end{array} \qquad \begin{array}{r} 3 \\ -1 \\ \hline 2 \end{array}$$

(b)

$$\begin{array}{r} 11 \\ -10 \\ \hline \mathbf{01} \end{array} \qquad \begin{array}{r} 3 \\ -2 \\ \hline 1 \end{array}$$

No borrows were required in this example. The binary number 01 is the same as 1.

Binary Subtraction

Subtract 011 from 101.

Solution

101	5
−011	−3
010	2

Left column:
When a 1 is borrowed,
a 0 is left, so 0 − 0 = 0.

Middle column:
Borrow 1 from next column
to the left, making a 10 in
this column, then 10 − 1 = 1.

$$\begin{array}{r} {}^{0}\,{}^{1}\,01 \\ -0\,11 \\ \hline 0\,10 \end{array}$$

Right column:
1 − 1 = 0

Binary Multiplication

The four basic rules for multiplying bits are as follows:

$$0 \times 0 = 0$$
$$0 \times 1 = 0$$
$$1 \times 0 = 0$$
$$1 \times 1 = 1$$

Binary Multiplication

Perform the following binary multiplications:

(a) 11×11 **(b)** 101×111

Solution

(a)

$$\begin{array}{r} 11 \\ \times\ 11 \\ \hline 11 \\ +11 \\ \hline \mathbf{1001} \end{array} \qquad \begin{array}{r} 3 \\ \times\ 3 \\ \hline 9 \end{array}$$

Partial products

(b)

$$\begin{array}{r} 111 \\ \times\ 101 \\ \hline 111 \\ 000 \\ +111 \\ \hline \mathbf{100011} \end{array} \qquad \begin{array}{r} 7 \\ \times\ 5 \\ \hline 35 \end{array}$$

Partial products

Binary Division

Perform the following binary divisions:

(a) $110 \div 11$ **(b)** $110 \div 10$

Solution

(a)

$$\begin{array}{r} \mathbf{10} \\ 11\overline{)110} \\ \underline{11} \\ 000 \end{array} \qquad \begin{array}{r} 2 \\ 3\overline{)6} \\ \underline{6} \\ 0 \end{array}$$

(b)

$$\begin{array}{r} \mathbf{11} \\ 10\overline{)110} \\ \underline{10} \\ 10 \\ \underline{10} \\ 00 \end{array} \qquad \begin{array}{r} 3 \\ 2\overline{)6} \\ \underline{6} \\ 0 \end{array}$$

5. Convert the following binary numbers to decimal:

(a) 001 (b) 010 (c) 101 (d) 110
(e) 1010 (f) 1011 (g) 1110 (h) 1111

6. Convert the following binary numbers into decimal:

(a) 100001 (b) 100111 (c) 101010 (d) 111001
(e) 1100000 (f) 11111101 (g) 11110010 (h) 11111111

7. Convert each binary number to decimal:

(a) 110011.11 (b) 101010.01 (c) 1000001.111
(d) 1111000.101 (e) 1011100.10101 (f) 1110001.0001
(g) 1011010.1010 (h) 1111111.11111

8. What is the highest decimal number that can be represented by each of the following numbers of binary digits (bits)?

(a) two (b) three (c) four (d) five (e) six
(f) seven (g) eight (h) nine (i) ten (j) eleven

9. How many bits are required to represent the following decimal numbers?

(a) 5 (b) 10 (c) 15 (d) 20
(e) 100 (f) 120 (g) 140 (h) 160

10. Generate the binary sequence for each decimal sequence:

(a) 0 through 7 (b) 8 through 15 (c) 16 through 31
(d) 32 through 63 (e) 64 through 75

15. Add the binary numbers:

(a) 10 + 10 **(b)** 10 + 11 **(c)** 100 + 11
(d) 111 + 101 **(e)** 1111 + 111 **(f)** 1111 + 1111

16. Use direct subtraction on the following binary numbers:

(a) 10 − 1 **(b)** 100 − 11 **(c)** 110 − 100
(d) 1111 − 11 **(e)** 1101 − 101 **(f)** 110000 − 1111

17. Perform the following binary multiplications:

(a) 11 × 10 **(b)** 101 × 11 **(c)** 111 × 110
(d) 1100 × 101 **(e)** 1110 × 1110 **(f)** 1111 × 1100

18. Divide the binary numbers as indicated:

(a) 110 ÷ 11 **(b)** 1010 ÷ 10 **(c)** 1111 ÷ 101

Complements of Binary Numbers

After completing this section, you should be able to

- Convert a binary number to its 1's complement
- Convert a binary number to its 2's complement using either of two methods

Finding the 1's Complement

The 1's **complement** of a binary number is found by changing all 1s to 0s and all 0s to 1s, as illustrated below:

Change each bit in a number to get the 1's complement.

1 0 1 1 0 0 1 0	Binary number
↓↓↓↓↓↓↓↓	
0 1 0 0 1 1 0 1	1's complement

Complements of Binary Numbers

Finding the 2's Complement

The 2's complement of a binary number is found by adding 1 to the LSB of the 1's complement.

$$2\text{'s complement} = (1\text{'s complement}) + 1$$

Find the 2's complement of 10110010.

Solution

	10110010	Binary number
	01001101	1's complement
+	1	Add 1
	01001110	2's complement

Complements of Binary Numbers

Finding the 2's Complement

1. Start at the right with the LSB and write the bits as they are up to and including the first 1.
2. Take the 1's complements of the remaining bits.

Find the 2's complement of 10111000 using the alternative method.

Solution

10111000	Binary number
01001000	2's complement

1's complements of original bits

These bits stay the same.

Signed Numbers

After completing this section, you should be able to

- Express positive and negative numbers in sign-magnitude
- Express positive and negative numbers in 1's complement
- Express positive and negative numbers in 2's complement
- Determine the decimal value of signed binary numbers
- Express a binary number in floating-point format

Signed Numbers

The Sign Bit

The left-most bit in a signed binary number is the **sign bit**, which tells you whether the number is positive or negative.

A 0 sign bit indicates a positive number, and a 1 sign bit indicates a negative number.

For example, the decimal number +25 is expressed as an 8-bit signed binary number using the sign-magnitude form as

$$0\underbrace{0011001}_{\text{Magnitude bits}}$$

Sign bit

The decimal number −25 is expressed as

$$10011001$$

Notice that the only difference between +25 and -25 is the sign bit because the magnitude bits are in true binary for both positive and negative numbers.

In the sign-magnitude form, a negative number has the same magnitude bits as the corresponding positive number but the sign bit is a 1 rather than a zero.

Signed Numbers

1's Complement Form

Positive numbers in 1's complement form are represented the same way as the positive sign-magnitude numbers. Negative numbers, however, are the 1's complements of the corresponding positive numbers. For example, using eight bits, the decimal number −25 is expressed as the 1's complement of +25 (00011001) as

$$11100110$$

In the 1's complement form, a negative number is the 1's complement of the corresponding positive number.

2's Complement Form

Positive numbers in 2's complement form are represented the same way as in the sign-magnitude and 1's complement forms. Negative numbers are the 2's complements of the corresponding positive numbers. Again, using eight bits, let's take decimal number −25 and express it as the 2's complement of +25 (00011001). Inverting each bit and adding 1, you get

$$-25 = 11100111$$

In the 2's complement form, a negative number is the 2's complement of the corresponding positive number.

Signed Numbers

Express the decimal number −39 as an 8-bit number in the sign-magnitude, 1's complement, and 2's complement forms.

Solution

First, write the 8-bit number for +39.

00100111

In the *sign-magnitude form,* −39 is produced by changing the sign bit to a 1 and leaving the magnitude bits as they are. The number is

10100111

In the *1's complement form,* −39 is produced by taking the 1's complement of +39 (00100111).

11011000

In the *2's complement form,* −39 is produced by taking the 2's complement of +39 (00100111) as follows:

$$\begin{array}{rl} 11011000 & \text{1's complement} \\ + \quad\quad\quad 1 & \\ \hline \mathbf{11011001} & \text{2's complement} \end{array}$$

Signed Numbers

The Decimal Value of Signed Numbers

Sign-Magnitude

Decimal values of positive and negative numbers in the sign-magnitude form are determined by summing the weights in all the magnitude bit positions where there are 1s and ignoring those positions where there are zeros. The sign is determined by examination of the sign bit.

Determine the decimal value of this signed binary number expressed in sign-magnitude: 10010101.

Solution

The seven magnitude bits and their powers-of-two weights are as follows:

2^6	2^5	2^4	2^3	2^2	2^1	2^0
0	0	1	0	1	0	1

Summing the weights where there are 1s,

$$16 + 4 + 1 = 21$$

The sign bit is 1; therefore, the decimal number is **−21**.

Signed Numbers

The Decimal Value of Signed Numbers

1's Complement

Determine the decimal values of the signed binary numbers expressed in 1's complement:

(a) 00010111 **(b)** 11101000

Solution

(a) The bits and their powers-of-two weights for the positive number are as follows:

-2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
0	0	0	1	0	1	1	1

Summing the weights where there are 1s,

$$16 + 4 + 2 + 1 = \mathbf{+23}$$

(b) The bits and their powers-of-two weights for the negative number are as follows. Notice that the negative sign bit has a weight of -2^7 or -128.

-2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1	1	1	0	1	0	0	0

Summing the weights where there are 1s,

$$-128 + 64 + 32 + 8 = -24$$

Adding 1 to the result, the final decimal number is

$$-24 + 1 = \mathbf{-23}$$

Signed Numbers

The Decimal Value of Signed Numbers

2's Complement

Determine the decimal values of the signed binary numbers expressed in 2's complement:

(a) 01010110 **(b)** 10101010

Solution

(a) The bits and their powers-of-two weights for the positive number are as follows:

-2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
0	1	0	1	0	1	1	0

Summing the weights where there are 1s,

$$64 + 16 + 4 + 2 = \mathbf{+86}$$

(b) The bits and their powers-of-two weights for the negative number are as follows. Notice that the negative sign bit has a weight of $-2^7 = -128$.

-2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1	0	1	0	1	0	1	0

Summing the weights where there are 1s,

$$-128 + 32 + 8 + 2 = \mathbf{-86}$$

Signed Numbers

21. Determine the 1's complement of each binary number:

(a) 100 (b) 111 (c) 1100
(d) 10111011 (e) 1001010 (f) 10101010

22. Determine the 2's complement of each binary number using either method:

(a) 11 (b) 110 (c) 1010 (d) 1001
(e) 101010 (f) 11001 (g) 11001100 (h) 11000111

Arithmetic Operations with Signed Numbers

Addition

The result is the sum. There are four cases that can occur when two signed binary numbers are added.

1. Both numbers positive
2. Positive number with magnitude larger than negative number
3. Negative number with magnitude larger than positive number
4. Both numbers negative

Arithmetic Operations with Signed Numbers

Addition

Both numbers positive:

00000111	7
+ 00000100	+ 4
00001011	11

The sum is positive and is therefore in true (uncomplemented) binary.

Addition of two positive numbers yields a positive number.

Positive number with magnitude larger than negative number:

	00001111	15
	+ 11111010	+ −6
Discard carry →	1 00001001	9

The final carry bit is discarded. The sum is positive and therefore in true (uncomplemented) binary.

Addition of a positive number and a smaller negative number yields a positive number.

Arithmetic Operations with Signed Numbers

Addition

Negative number with magnitude larger than positive number:

Addition of a positive number and a larger negative number or two negative numbers yields a negative number in 2's complement.

$$\begin{array}{r} 00010000 \\ +\ 11101000 \\ \hline 11111000 \end{array} \qquad \begin{array}{r} 16 \\ +\ -24 \\ \hline -8 \end{array}$$

The sum is negative and therefore in 2's complement form.

Both numbers negative:

$$\begin{array}{rr} & 11111011 \\ & +\ 11110111 \\ \hline \text{Discard carry} \longrightarrow 1 & 11110010 \end{array} \qquad \begin{array}{r} -5 \\ +\ -9 \\ \hline -14 \end{array}$$

The final carry bit is discarded. The sum is negative and therefore in 2's complement form.

Arithmetic Operations with Signed Numbers

Numbers added two at a time

Add the signed numbers: 01000100, 00011011, 00001110, and 00010010.

Solution

The equivalent decimal additions are given for reference.

68	01000100	
+ 27	+ 00011011	Add 1st two numbers
95	01011111	1st sum
+ 14	+ 00001110	Add 3rd number
109	01101101	2nd sum
+ 18	+ 00010010	Add 4th number
127	**01111111**	Final sum

Arithmetic Operations with Signed Numbers

Subtraction

Perform each of the following subtractions of the signed numbers:

(a) 00001000 − 00000011 **(b)** 00001100 − 11110111

(c) 11100111 − 00010011 **(d)** 10001000 − 11100010

Solution

Like in other examples, the equivalent decimal subtractions are given for reference.

(a) In this case, $8 - 3 = 8 + (-3) = 5$.

	00001000	Minuend (+8)
	+ 11111101	2's complement of subtrahend (−3)
Discard carry →	**1 00000101**	Difference (+5)

(b) In this case, $12 - (-9) = 12 + 9 = 21$.

00001100	Minuend (+12)
+ 00001001	2's complement of subtrahend (+9)
00010101	Difference (+21)

Arithmetic Operations with Signed Numbers

Subtraction

Perform each of the following subtractions of the signed numbers:

(a) 00001000 − 00000011 (b) 00001100 − 11110111

(c) 11100111 − 00010011 (d) 10001000 − 11100010

(c) In this case, $-25 - (+19) = -25 + (-19) = -44$.

	11100111	Minuend (−25)
	+ 11101101	2's complement of subtrahend (−19)
Discard carry	**1 11010100**	Difference (−44)

(d) In this case, $-120 - (-30) = -120 + 30 = -90$.

10001000	Minuend (−120)
+ 00011110	2's complement of subtrahend (+30)
10100110	Difference (−90)

Arithmetic Operations with Signed Numbers

23. Express each decimal number in binary as an 8-bit sign-magnitude number:

(a) +29 **(b)** −85 **(c)** +100 **(d)** −123

24. Express each decimal number as an 8-bit number in the 1's complement form:

(a) −34 **(b)** +57 **(c)** −99 **(d)** +115

25. Express each decimal number as an 8-bit number in the 2's complement form:

(a) +12 **(b)** −68 **(c)** +101 **(d)** −125

26. Determine the decimal value of each signed binary number in the sign-magnitude form:

(a) 10011001 **(b)** 01110100 **(c)** 10111111

27. Determine the decimal value of each signed binary number in the 1's complement form:

(a) 10011001 **(b)** 01110100 **(c)** 10111111

28. Determine the decimal value of each signed binary number in the 2's complement form:

(a) 10011001 **(b)** 01110100 **(c)** 10111111

Hexadecimal Numbers

After completing this section, you should be able to List the hexadecimal characters

- Count in hexadecimal
- Convert from binary to hexadecimal
- Convert from hexadecimal to binary
- Convert from hexadecimal to decimal
- Convert from decimal to hexadecimal
- Add hexadecimal numbers
- Determine the 2's complement of a hexadecimal number
- Subtract hexadecimal number

Hexadecimal Numbers

The hexadecimal number system has a base of sixteen; that is, it is composed of 16 numeric and alphabetic characters. Most digital systems process binary data in groups that are multiples of four bits, making the hexadecimal number very convenient because each hexadecimal digit represents a 4-bit binary number

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Hexadecimal Numbers

Binary-to-Hexadecimal Conversion

Convert the following binary numbers to hexadecimal:

(a) 1100101001010111 **(b)** 111111000101101001

Solution

(a) $\underbrace{1100}_{C}\underbrace{1010}_{A}\underbrace{0101}_{5}\underbrace{0111}_{7} = \mathbf{CA57}_{16}$

(b) $\underbrace{0011}_{3}\underbrace{1111}_{F}\underbrace{0001}_{1}\underbrace{0110}_{6}\underbrace{1001}_{9} = \mathbf{3F169}_{16}$

Two zeros have been added in part (b) to complete a 4-bit group at the left.

Hexadecimal Numbers

Hexadecimal-to-Binary Conversion

Determine the binary numbers for the following hexadecimal numbers:

(a) $10A4_{16}$ (b) $CF8E_{16}$ (c) 9742_{16}

Solution

(a) 1 0 A 4

$$\underbrace{1}\,\underbrace{000}\,\underbrace{1010}\,\underbrace{0100}$$

(b) C F 8 E

$$\underbrace{1100}\,\underbrace{1111}\,\underbrace{1000}\,\underbrace{1110}$$

(c) 9 7 4 2

$$\underbrace{1001}\,\underbrace{0111}\,\underbrace{0100}\,\underbrace{0010}$$

In part (a), the MSB is understood to have three zeros preceding it, thus forming a 4-bit group.

Hexadecimal Numbers

Hexadecimal-to-Decimal Conversion

Convert the following hexadecimal numbers to decimal:

(a) $1C_{16}$ **(b)** $A85_{16}$

Solution

Remember, convert the hexadecimal number to binary first, then to decimal.

(a)

$$\overbrace{0001}^{1\,\downarrow}\overbrace{1100}^{C\,\downarrow} = 2^4 + 2^3 + 2^2 = 16 + 8 + 4 = \mathbf{28}_{10}$$

(b)

$$\overbrace{1010}^{A\,\downarrow}\overbrace{1000}^{8\,\downarrow}\overbrace{0101}^{5\,\downarrow} = 2^{11} + 2^9 + 2^7 + 2^2 + 2^0 = 2048 + 512 + 128 + 4 + 1 = \mathbf{2693}_{10}$$

Hexadecimal Numbers

Hexadecimal-to-Decimal Conversion

Another way to convert a hexadecimal number to its decimal equivalent is to multiply the decimal value of each hexadecimal digit by its weight and then take the sum of these products. The weights of a hexadecimal number are increasing powers of 16 (from right to left). For a 4-digit hexadecimal number, the weights are

16^3	16^2	16^1	16^0
4096	256	16	1

Hexadecimal Numbers

Hexadecimal-to-Decimal Conversion

Convert the following hexadecimal numbers to decimal:

(a) $E5_{16}$ **(b)** $B2F8_{16}$

Solution

Recall from Table 2–3 that letters A through F represent decimal numbers 10 through 15, respectively.

(a) $E5_{16} = (E \times 16) + (5 \times 1) = (14 \times 16) + (5 \times 1) = 224 + 5 = \mathbf{229}_{10}$

(b)
$$\begin{aligned} B2F8_{16} &= (B \times 4096) + (2 \times 256) + (F \times 16) + (8 \times 1) \\ &= (11 \times 4096) + (2 \times 256) + (15 \times 16) + (8 \times 1) \\ &= 45{,}056 + 512 + 240 + 8 = \mathbf{45{,}816}_{10} \end{aligned}$$

Hexadecimal Numbers

Decimal-to-Hexadecimal Conversion

Convert the decimal number 650 to hexadecimal by repeated division by 16.

Solution

Hexadecimal remainder

$$\frac{650}{16} = 40.625 \rightarrow 0.625 \times 16 = 10 = \quad A$$

$$\frac{40}{16} = 2.5 \longrightarrow 0.5 \times 16 = 8 = \quad 8$$

$$\frac{2}{16} = 0.125 \longrightarrow 0.125 \times 16 = 2 = \quad 2$$

Stop when whole number quotient is zero.

2 8 A Hexadecimal number

MSD LSD

Hexadecimal Numbers

Hexadecimal Addition

Add the following hexadecimal numbers:

(a) $23_{16} + 16_{16}$ (b) $58_{16} + 22_{16}$ (c) $2B_{16} + 84_{16}$ (d) $DF_{16} + AC_{16}$

Solution

(a)

$$\begin{array}{r} 23_{16} \\ +\ 16_{16} \\ \hline \mathbf{39_{16}} \end{array}$$

right column: $3_{16} + 6_{16} = 3_{10} + 6_{10} = 9_{10} = 9_{16}$

left column: $2_{16} + 1_{16} = 2_{10} + 1_{10} = 3_{10} = 3_{16}$

Hexadecimal Numbers

Hexadecimal Addition

(b)	58_{16}	right column:	$8_{16} + 2_{16} = 8_{10} + 2_{10} = 10_{10} = A_{16}$
	$+\ 22_{16}$	left column:	$5_{16} + 2_{16} = 5_{10} + 2_{10} = 7_{10} = 7_{16}$
	$\mathbf{7A_{16}}$		
(c)	$2B_{16}$	right column:	$B_{16} + 4_{16} = 11_{10} + 4_{10} = 15_{10} = F_{16}$
	$+\ 84_{16}$	left column:	$2_{16} + 8_{16} = 2_{10} + 8_{10} = 10_{10} = A_{16}$
	$\mathbf{AF_{16}}$		
(d)	DF_{16}	right column:	$F_{16} + C_{16} = 15_{10} + 12_{10} = 27_{10}$
	$+\ AC_{16}$		$27_{10} - 16_{10} = 11_{10} = B_{16}$ with a 1 carry
	$\mathbf{18B_{16}}$	left column:	$D_{16} + A_{16} + 1_{16} = 13_{10} + 10_{10} + 1_{10} = 24_{10}$
			$24_{10} - 16_{10} = 8_{10} = 8_{16}$ with a 1 carry

Hexadecimal Numbers

Hexadecimal Subtraction

Subtract the following hexadecimal numbers:

(a) $84_{16} - 2A_{16}$ **(b)** $C3_{16} - 0B_{16}$

Solution

(a) $2A_{16} = 00101010$

2's complement of $2A_{16} = 11010110 = D6_{16}$ (using Method 1)

$$\begin{array}{r} 84_{16} \\ +\ D6_{16} \\ \hline \cancel{1}5A_{16} \end{array}$$

Add

Drop carry, as in 2's complement addition

The difference is $\mathbf{5A_{16}}$.

(b) $0B_{16} = 00001011$

2's complement of $0B_{16} = 11110101 = F5_{16}$ (using Method 1)

$$\begin{array}{r} C3_{16} \\ +\ F5_{16} \\ \hline \cancel{1}B8_{16} \end{array}$$

Add

Drop carry

The difference is $\mathbf{B8_{16}}$.

Hexadecimal Numbers

37. Convert each hexadecimal number to binary:

(a) 46_{16} **(b)** 54_{16} **(c)** $B4_{16}$ **(d)** $1A3_{16}$
(e) FA_{16} **(f)** ABC_{16} **(g)** $ABCD_{16}$

38. Convert each binary number to hexadecimal:

(a) 1111 **(b)** 1011 **(c)** 11111
(d) 10101010 **(e)** 10101100 **(f)** 10111011

39. Convert each hexadecimal number to decimal:

(a) 42_{16} **(b)** 64_{16} **(c)** $2B_{16}$ **(d)** $4D_{16}$
(e) FF_{16} **(f)** BC_{16} **(g)** $6F1_{16}$ **(h)** ABC_{16}

40. Convert each decimal number to hexadecimal:

(a) 10 **(b)** 15 **(c)** 32 **(d)** 54
(e) 365 **(f)** 3652 **(g)** 7825 **(h)** 8925

41. Perform the following additions:

(a) $25_{16} + 33_{16}$ **(b)** $43_{16} + 62_{16}$ **(c)** $A4_{16} + F5_{16}$ **(d)** $FC_{16} + AE_{16}$

42. Perform the following subtractions:

(a) $60_{16} - 39_{16}$ **(b)** $A5_{16} - 98_{16}$ **(c)** $F1_{16} - A6_{16}$ **(d)** $AC_{16} - 10_{16}$

Octal Numbers

After completing this section, you should be able to

- Write the digits of the octal number system
- Convert from octal to decimal
- Convert from decimal to octal
- Convert from octal to binary
- Convert from binary to octal

Octal Numbers

The octal number system is composed of eight digits, which are

0, 1, 2, 3, 4, 5, 6, 7

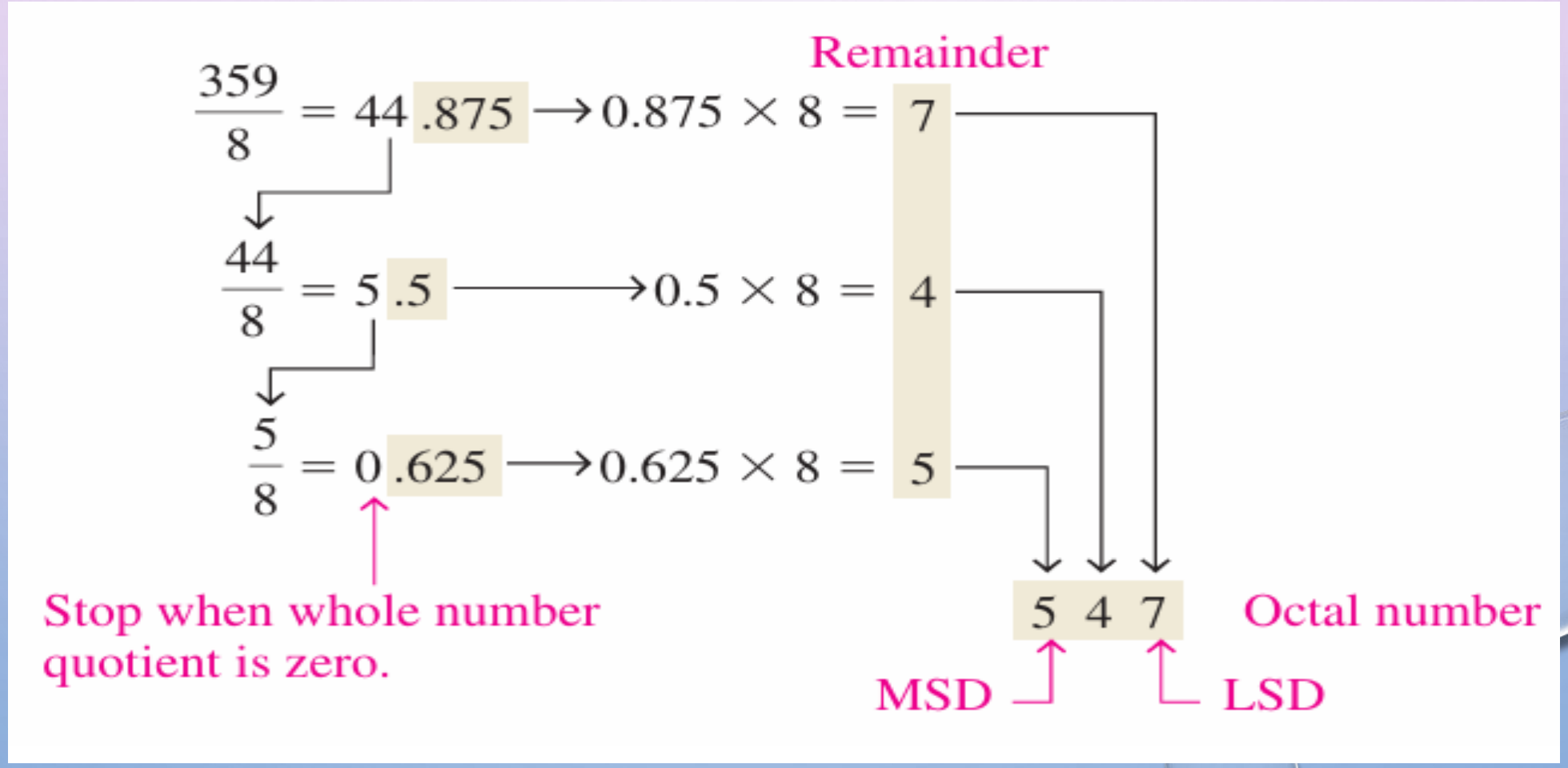
Octal-to-Decimal Conversion

$$\begin{array}{rl} \text{Weight:} & 8^3\ 8^2\ 8^1\ 8^0 \\ \text{Octal number:} & 2\ \ 3\ \ 7\ \ 4 \end{array}$$

$$\begin{aligned} 2374_8 &= (2 \times 8^3) + (3 \times 8^2) + (7 \times 8^1) + (4 \times 8^0) \\ &= (2 \times 512) + (3 \times 64) + (7 \times 8) + (4 \times 1) \\ &= 1024 + 192 + 56 + 4 = 1276_{10} \end{aligned}$$

Octal Numbers

Decimal-to-Octal Conversion



Octal Numbers

Binary-to-Octal Conversion

Octal/binary conversion.

Octal Digit	0	1	2	3	4	5	6	7
Binary	000	001	010	011	100	101	110	111

Convert each of the following binary numbers to octal:

(a) 110101 (b) 101111001 (c) 100110011010 (d) 11010000100

Solution

(a) $\underbrace{110}_{6}\underbrace{101}_{5} = \mathbf{65}_8$

(b) $\underbrace{101}_{5}\underbrace{111}_{7}\underbrace{001}_{1} = \mathbf{571}_8$

(c) $\underbrace{100}_{4}\underbrace{110}_{6}\underbrace{011}_{3}\underbrace{010}_{2} = \mathbf{4632}_8$

(d) $\underbrace{011}_{3}\underbrace{010}_{2}\underbrace{000}_{0}\underbrace{100}_{4} = \mathbf{3204}_8$

Octal Numbers

43. Convert each octal number to decimal:

(a) 14_8 **(b)** 53_8 **(c)** 67_8 **(d)** 174_8
(e) 635_8 **(f)** 254_8 **(g)** 2673_8 **(h)** 7777_8

44. Convert each decimal number to octal by repeated division by 8:

(a) 23 **(b)** 45 **(c)** 65 **(d)** 84
(e) 124 **(f)** 156 **(g)** 654 **(h)** 9999

45. Convert each octal number into binary:

(a) 17_8 **(b)** 26_8 **(c)** 145_8 **(d)** 456_8
(e) 653_8 **(f)** 777_8

46. Convert each binary number to octal:

(a) 100 **(b)** 110 **(c)** 1100
(d) 1111 **(e)** 11001 **(f)** 11110
(g) 110011 **(h)** 101010 **(i)** 10101111

Thank You